

AIIMS CRE – OT/Anaesthesia Technician

Comprehensive Practice Question Bank — 100 Original MCQs with Explanations

These are **original practice questions**, written to match the topic spread, style, and difficulty of the AIIMS CRE OT Technician previous-year paper you shared (not verbatim leaked questions — no verified official bank exists for this post, so treat this as high-yield revision material alongside your official study sources). In the actual AIIMS CRE, the domain-specific paper is 60 of the 100 total questions, alongside 20 General Knowledge/Aptitude and 20 Computer Knowledge questions — this set focuses entirely on the OT/Anaesthesia domain portion.

Section A — Operation Theatre Technology, Sterilization & Instruments (Q1–Q25)

Q1. The operation theatre complex is conventionally divided into how many zones based on the level of asepsis maintained? A) 2 B) 3 C) 4 D) 5

Answer: C) 4 **Explanation:** The OT complex is divided into four zones — Protective (reception/transfer), Clean (changing rooms, offices), Sterile/Aseptic (operating rooms), and Disposal (dirty utility) — with bacterial count progressively decreasing from the outer to the inner zone.

Q2. The ideal temperature to be maintained inside a modern operating theatre is: A) 15–18°C B) 20–24°C C) 26–30°C D) 32–35°C

Answer: B) 20–24°C **Explanation:** An OT temperature of 20–24°C balances patient thermoregulation with staff comfort and limits bacterial proliferation; it is typically paired with a relative humidity of 50–60%.

Q3. Laminar air flow systems in the OT are primarily installed to: A) Increase room temperature B) Reduce airborne microbial contamination at the surgical site C) Reduce noise levels D) Improve lighting

Answer: B) Reduce airborne microbial contamination at the surgical site **Explanation:** Laminar flow delivers unidirectional, HEPA-filtered air over the operating field, sweeping particulate and microbial contaminants away from the wound — most valuable in high infection-risk procedures like joint replacements.

Q4. HEPA filters used in OT ventilation remove particles of size: A) ≥ 0.3 microns with 99.97% efficiency B) ≥ 3 microns with 90% efficiency C) ≥ 30 microns with 99% efficiency D) ≥ 0.03 microns with 100% efficiency

Answer: A) ≥ 0.3 microns with 99.97% efficiency **Explanation:** HEPA filters are rated to capture

at least 99.97% of particles ≥ 0.3 microns — the "most penetrating particle size" — making them effective against bacteria and most airborne pathogens.

Q5. The standard steam autoclave sterilization cycle operates at: A) 100°C for 30 minutes B) 121°C at 15 psi for 15–20 minutes C) 160°C for 2 hours D) 90°C for 60 minutes

Answer: B) 121°C at 15 psi for 15–20 minutes **Explanation:** Standard gravity-displacement autoclaving uses saturated steam at 121°C/15 psi for 15–20 minutes; higher-pressure "flash" cycles use 134°C for shorter durations for urgently needed unwrapped instruments.

Q6. Ethylene oxide (ETO) sterilization is preferred for: A) Metal surgical instruments B) Heat- and moisture-sensitive items like endoscopes and plastics C) Linen and drapes D) Glassware

Answer: B) Heat- and moisture-sensitive items like endoscopes and plastics **Explanation:** ETO is a low-temperature gas sterilant (37–63°C) suited to heat-labile items; because it is toxic and carcinogenic, sterilized items require a mandatory aeration period before use.

Q7. The Bowie-Dick test is used to check: A) Efficiency of air removal in a pre-vacuum autoclave B) Sharpness of surgical blades C) Concentration of glutaraldehyde D) Functioning of the anaesthesia ventilator

Answer: A) Efficiency of air removal in a pre-vacuum autoclave **Explanation:** The Bowie-Dick test detects residual air pockets that would prevent steam penetration in high-vacuum autoclaves; it is run daily before the first sterilization load.

Q8. Cold sterilization of heat-sensitive instruments is commonly performed using: A) 2% Glutaraldehyde B) Normal saline C) 70% alcohol only D) Boiling water

Answer: A) 2% Glutaraldehyde **Explanation:** 2% glutaraldehyde (e.g., Cidex) achieves high-level disinfection with 20–30 minutes of contact and sterilization with prolonged immersion (~10 hours), making it a standard chemical sterilant for heat-sensitive equipment.

Q9. CSSD stands for: A) Central Sterile Supply Department B) Clinical Surgical Store Department C) Central Surgical Safety Division D) Combined Sterilization and Storage Depot

Answer: A) Central Sterile Supply Department **Explanation:** CSSD is the hospital department responsible for collecting, cleaning, disinfecting, packaging, sterilizing, and distributing reusable surgical instruments and supplies to all departments.

Q10. As per India's Biomedical Waste Management Rules, human anatomical waste and soiled waste are segregated into: A) Red bag B) Yellow bag C) Blue bag D) White (translucent) container

Answer: B) Yellow bag **Explanation:** Yellow bags/containers are designated for human

anatomical waste, soiled waste, expired medicines, and chemical waste, which are sent for incineration (or deep burial where permitted).

Q11. Contaminated recyclable waste such as IV tubing, catheters, and (emptied) blood/urine bags is collected in: A) Yellow bag B) Red bag C) White container D) Blue bag

Answer: B) Red bag **Explanation:** Red bags are used for contaminated plastic waste (tubing, catheters, IV sets without needles) that is autoclaved or microwaved and then shredded prior to recycling.

Q12. Needles and other sharps are disposed of in: A) Red bag B) Yellow bag C) White (translucent), puncture-proof containers D) Blue bag

Answer: C) White (translucent), puncture-proof containers **Explanation:** Sharps require puncture-proof, leak-proof white/translucent containers to prevent needle-stick injuries; contents are treated by disinfection/autoclaving followed by shredding or encapsulation. (Blue containers, by contrast, are reserved for glassware and metallic implants.)

Q13. The first surgical hand scrub of the operating day should ideally last: A) 1 minute B) 5 minutes C) 15 minutes D) 30 seconds

Answer: B) 5 minutes **Explanation:** Traditional teaching specifies a 5-minute scrub for the first case of the day, with subsequent scrubs between cases shortened to about 3 minutes if hands remain visibly clean.

Q14. Which scalpel blade number is most commonly used to make a stab/puncture incision? A) No. 10 B) No. 11 C) No. 15 D) No. 22

Answer: B) No. 11 **Explanation:** The pointed No. 11 blade is designed for stab incisions (e.g., abscess drainage), while No. 10, 15, and 22 are curved blades suited to longer skin incisions.

Q15. Which forceps is preferred for handling bowel and other delicate tissue due to its atraumatic grip? A) Allis forceps B) Babcock forceps C) Kocher forceps D) Artery forceps

Answer: B) Babcock forceps **Explanation:** Babcock forceps have broad, fenestrated, non-crushing jaws that grasp delicate structures like bowel without causing trauma, unlike the toothed Allis or crushing Kocher forceps.

Q16. Mayo scissors are primarily designed for: A) Cutting sutures and tough tissue B) Fine dissection of delicate tissue C) Cutting dressings only D) Cutting bone

Answer: A) Cutting sutures and tough tissue **Explanation:** Mayo scissors have shorter, sturdier blades suited to cutting sutures and heavier tissue, while the thinner, angled Metzenbaum scissors are reserved for delicate tissue dissection.

Q17. Which of the following is an absorbable suture material? A) Silk B) Polypropylene (Prolene) C) Polyglactin (Vicryl) D) Nylon

Answer: C) Polyglactin (Vicryl) **Explanation:** Vicryl (polyglactin 910) is a synthetic absorbable suture broken down by hydrolysis over weeks, whereas silk, prolene, and nylon are non-absorbable materials that remain in tissue unless removed.

Q18. A 6-0 suture is _____ than a 2-0 suture. A) Thicker B) Thinner C) The same size D) Not comparable

Answer: B) Thinner **Explanation:** On the USP suture scale, a higher number of zeros denotes a progressively finer suture; 6-0 is finer than 2-0 and is typically used for delicate work like vascular or ophthalmic surgery.

Q19. In monopolar electrocautery, the return (grounding) pad is necessary because it: A) Increases current output B) Completes the electrical circuit safely back to the generator C) Sterilizes the tissue D) Is not actually necessary

Answer: B) Completes the electrical circuit safely back to the generator **Explanation:** Monopolar diathermy passes current from the active electrode through the patient's body to a dispersive return pad, safely completing the circuit; bipolar cautery needs no return pad since current passes only between the two forceps tips.

Q20. The three elements of the "surgical fire triad" in the OT are an oxidizer, a fuel source, and: A) A humidifier B) An ignition source C) A cooling agent D) A sterilant

Answer: B) An ignition source **Explanation:** Surgical fires require an oxidizer (oxygen/nitrous oxide-enriched atmosphere), fuel (drapes, alcohol-based skin preps), and an ignition source (electrocautery, laser) — removing any one element prevents fire.

Q21. Sponge and instrument counts in the OT are primarily performed to: A) Speed up the surgery B) Prevent retained surgical items in the patient C) Reduce the cost of surgery D) Assist the anaesthetist

Answer: B) Prevent retained surgical items in the patient **Explanation:** Counts are performed jointly by scrub and circulating staff before the procedure, before cavity closure, and at skin closure, to ensure nothing is inadvertently left inside the patient.

Q22. A self-retaining abdominal wall retractor commonly used in laparotomy is the: A) Langenbeck retractor B) Balfour retractor C) Deaver retractor D) Army-Navy retractor

Answer: B) Balfour retractor **Explanation:** The Balfour retractor is self-retaining and hand-free, holding abdominal wall edges apart during laparotomy, unlike hand-held retractors such as Langenbeck, Deaver, or Army-Navy.

Q23. Biological indicators used to confirm sterilization efficacy in an autoclave typically use spores of: A) *Staphylococcus aureus* B) *Geobacillus stearothermophilus* C) *Escherichia coli* D) *Candida albicans*

Answer: B) *Geobacillus stearothermophilus* **Explanation:** These spores are highly resistant to steam sterilization, making them the standard biological indicator to confirm that an autoclave cycle actually achieved sterilizing conditions, not just the correct temperature/time settings.

Q24. The recommended number of total air changes per hour in a conventional operating theatre is approximately: A) 5–8 B) 20–25 C) 40–50 D) 60–70

Answer: B) 20–25 **Explanation:** Standard OT ventilation guidelines recommend roughly 20–25 total air changes per hour, with a proportion of fresh air, to dilute airborne contaminants and maintain positive pressure relative to adjacent areas.

Q25. Which property is essential in surgical gowns and drapes to prevent "strike-through" contamination? A) Bright color B) Fluid resistance/barrier property C) Light weight only D) Transparency

Answer: B) Fluid resistance/barrier property **Explanation:** Gowns and drapes must act as an effective barrier to fluid and microbial strike-through, especially over critical zones like the chest and forearms, to protect both patient and surgical team.

Section B — Anaesthesia Equipment & Airway Management (Q26–Q45)

Q26. The basic components of a modern anaesthesia workstation include all of the following EXCEPT: A) Gas supply and flowmeters B) Vaporizer C) Breathing circuit D) Defibrillator

Answer: D) Defibrillator **Explanation:** An anaesthesia machine comprises gas supply, flowmeters, vaporizer, breathing circuit, and often an integrated ventilator and scavenging system; a defibrillator is separate emergency equipment kept nearby, not part of the machine itself.

Q27. The continuous-flow anaesthesia machine is historically named after: A) Macintosh B) Boyle C) Magill D) Mapleson

Answer: B) Boyle **Explanation:** The "Boyle's machine," designed by Henry Boyle in the early 20th century, was the original continuous-flow apparatus and remains the eponym for modern anaesthesia workstations.

Q28. A specially heated, pressurized vaporizer (e.g., Tec 6) is required for which agent, due to its low boiling point? A) Sevoflurane B) Isoflurane C) Desflurane D) Halothane

Answer: C) Desflurane **Explanation:** Desflurane boils at $\sim 23.5^{\circ}\text{C}$, close to room temperature, so it requires an electrically heated, pressurized vaporizer to deliver a consistent concentration, unlike conventional variable-bypass vaporizers used for other agents.

Q29. The pin-index safety system is designed to prevent: A) Overfilling of cylinders B) Accidental connection of the wrong gas cylinder to the machine C) Leakage from the vaporizer D) Incorrect flowmeter readings

Answer: B) Accidental connection of the wrong gas cylinder to the machine **Explanation:** Each medical gas cylinder valve has a unique pin arrangement that matches a corresponding yoke on the machine, physically preventing it from being mounted in the wrong position.

Q30. The flowmeter (rotameter) used in anaesthesia machines works on the principle of: A) Fixed orifice, variable pressure B) Variable orifice, constant pressure differential C) Venturi effect only D) Bernoulli's principle only

Answer: B) Variable orifice, constant pressure differential **Explanation:** Gas flow raises a bobbin/float in a tapered tube; as it rises, the annular opening widens until the pressure differential across it balances the float's weight, so its position indicates the flow rate.

Q31. Which Mapleson breathing circuit is considered most efficient for spontaneous ventilation? A) Mapleson A (Magill circuit) B) Mapleson B C) Mapleson D D) Mapleson E

Answer: A) Mapleson A (Magill circuit) **Explanation:** The Mapleson A arrangement is most efficient during spontaneous breathing because exhaled gas is preferentially vented before fresh gas mixes with it; Mapleson D (Bain circuit) is instead most efficient for controlled ventilation.

Q32. The Bain circuit is a coaxial modification of which Mapleson system? A) Mapleson A B) Mapleson B C) Mapleson C D) Mapleson D

Answer: D) Mapleson D **Explanation:** The Bain circuit places the fresh gas inflow tube coaxially inside the corrugated expiratory limb of a Mapleson D circuit, making it lightweight while retaining Mapleson D's efficiency for controlled ventilation.

Q33. Which of the following is NOT a standard component of the circle absorber breathing system? A) Unidirectional valves B) CO₂ absorber canister C) Adjustable Pressure-Limiting (APL) valve D) Pin-index yoke

Answer: D) Pin-index yoke **Explanation:** The pin-index yoke belongs to the gas cylinder mounting on the machine, not the circle breathing system, which instead consists of inspiratory/expiratory valves, a CO₂ absorber, an APL valve, tubing, and a reservoir bag.

Q34. The internal diameter (mm) of an appropriately sized uncuffed paediatric endotracheal tube can be estimated using Cole's formula: A) $\text{Age}/2 + 12$ B) $\text{Age}/4 + 4$ C) $\text{Age} + 10$ D) $\text{Age}/3$

+ 5

Answer: B) Age/4 + 4 **Explanation:** Cole's formula (Age/4 + 4) estimates ETT internal diameter in children over 2 years — for a 4-year-old this predicts a 5.0 mm tube, always confirmed clinically with tubes half a size above and below available.

Q35. The small side hole near the tip of an endotracheal tube, allowing ventilation even if the main lumen abuts the tracheal wall, is called the: A) Murphy eye B) Magill orifice C) Sellick vent D) Cuff port

Answer: A) Murphy eye **Explanation:** The Murphy eye is a safety side-hole near the ETT tip that maintains gas flow even if the primary opening is occluded against the tracheal wall.

Q36. Magill forceps are primarily used to: A) Measure blood pressure B) Guide the endotracheal tube or retrieve a foreign body during nasal intubation/laryngoscopy C) Clamp blood vessels D) Retract the abdominal wall

Answer: B) Guide the endotracheal tube or retrieve a foreign body during nasal intubation/laryngoscopy **Explanation:** Angled Magill forceps let the operator direct the tip of a nasotracheal tube into the larynx under direct laryngoscopic vision, or retrieve foreign bodies from the oropharynx.

Q37. Compared to a Macintosh blade, a Miller (straight) laryngoscope blade is designed to: A) Sit in the vallecula and indirectly lift the epiglottis B) Directly lift the epiglottis C) Only be used in adults D) Provide no view of the vocal cords

Answer: B) Directly lift the epiglottis **Explanation:** The straight Miller blade tip passes beneath and directly lifts the epiglottis, whereas the curved Macintosh blade tip sits in the vallecula and elevates the epiglottis indirectly via the hyoepiglottic ligament.

Q38. A cricothyroidotomy is an emergency airway procedure performed through the: A) Cricothyroid membrane B) Thyroid gland C) 2nd–4th tracheal rings D) Hyoid bone

Answer: A) Cricothyroid membrane **Explanation:** Cricothyroidotomy provides rapid surgical airway access through the cricothyroid membrane in "cannot intubate, cannot ventilate" emergencies, and is quicker than a formal tracheostomy in that setting.

Q39. The Mallampati classification is used to assess: A) Depth of anaesthesia B) Predicted difficulty of laryngoscopy/intubation C) Degree of neuromuscular blockade D) Cardiac risk before surgery

Answer: B) Predicted difficulty of laryngoscopy/intubation **Explanation:** With the patient seated, mouth open, and tongue protruded, visible oropharyngeal structures are graded Class I–IV; higher classes correlate with greater likelihood of difficult intubation.

Q40. At an oxygen flow of 4 L/min via nasal cannula, the approximate FiO₂ delivered is: A) 24% B) 28% C) 36% D) 44%

Answer: C) 36% **Explanation:** A commonly taught rule of thumb adds roughly 4% FiO₂ per litre of nasal cannula flow above room air (21%): approximately 24% at 1 L/min, rising to about 36% at 4 L/min and 44% at 6 L/min.

Q41. Which LMA design has an additional channel allowing a gastric tube to be passed to decompress the stomach? A) Classic LMA B) ProSeal LMA C) Fastrach (Intubating) LMA D) LMA Unique

Answer: B) ProSeal LMA **Explanation:** The ProSeal LMA has a drainage tube running parallel to the airway tube, allowing gastric contents to be suctioned, which reduces (though does not eliminate) aspiration risk compared to the Classic LMA.

Q42. A significant limitation of the LMA compared to a cuffed endotracheal tube is that it: A) Cannot be used for spontaneous ventilation B) Does not reliably protect the airway from aspiration of gastric contents C) Cannot be inserted without a laryngoscope D) Is only available in one size

Answer: B) Does not reliably protect the airway from aspiration of gastric contents **Explanation:** Because the LMA cuff seals around the laryngeal inlet rather than within the trachea, it does not protect against regurgitation/aspiration as reliably as a cuffed ETT, making it unsuitable as first choice in a full-stomach patient.

Q43. Nitrogen gas cylinders, used to power pneumatic surgical tools, are conventionally colour-coded: A) Black B) French Blue C) Grey D) White

Answer: A) Black **Explanation:** Medical/industrial nitrogen cylinders are conventionally coded black, distinguishing them from oxygen, nitrous oxide (French Blue), and carbon dioxide (grey) cylinders on the same manifold.

Q44. Entonox, a pre-mixed cylinder of 50% nitrous oxide and 50% oxygen, is identified by which cylinder shoulder design? A) Blue and white quartered B) All black C) All white D) Yellow

Answer: A) Blue and white quartered **Explanation:** Entonox cylinders carry a blue-and-white quartered shoulder, visually distinguishing the premixed analgesic gas from plain nitrous oxide (French Blue) or oxygen cylinders.

Q45. The scavenging system on an anaesthesia machine is designed to: A) Increase fresh gas flow B) Remove waste anaesthetic gases from the OT environment C) Warm inspired gases D) Filter bacteria from the breathing circuit

Answer: B) Remove waste anaesthetic gases from the OT environment **Explanation:** Scavenging systems collect excess anaesthetic gas vented from the APL valve or ventilator and route it outside

the building, protecting OT staff from chronic occupational exposure to trace anaesthetic gases.

Section C — Anaesthesia Drugs & Pharmacology (Q46–Q65)

Q46. Which IV induction agent is classically described as a milky-white emulsion, often associated with profound amnesia? A) Thiopentone B) Etomidate C) Propofol D) Ketamine

Answer: C) Propofol **Explanation:** Propofol is formulated as a white lipid emulsion; it produces rapid, smooth induction and amnesia but commonly causes pain on injection and dose-dependent hypotension, and has no analgesic action of its own.

Q47. Thiopentone sodium is contraindicated in patients with: A) Bronchial asthma B) Porphyria C) Diabetes mellitus D) Hypertension

Answer: B) Porphyria **Explanation:** Thiopentone, a barbiturate, can precipitate a life-threatening acute attack in porphyria by inducing hepatic aminolevulinic acid synthetase; propofol or etomidate are generally preferred instead.

Q48. Which induction agent is preferred in haemodynamically unstable patients because it produces minimal cardiovascular depression? A) Thiopentone B) Propofol C) Etomidate D) Midazolam

Answer: C) Etomidate **Explanation:** Etomidate maintains cardiovascular stability better than thiopentone or propofol, making it useful for induction in shocked or cardiac-compromised patients, though it can cause adrenocortical suppression and injection pain/myoclonus.

Q49. Ketamine produces its anaesthetic effect mainly by: A) GABA receptor agonism B) NMDA receptor antagonism C) Sodium channel blockade D) Opioid receptor agonism

Answer: B) NMDA receptor antagonism **Explanation:** Ketamine blocks NMDA glutamate receptors, producing a unique "dissociative anaesthesia" state with profound analgesia while often preserving airway reflexes and spontaneous respiration.

Q50. Succinylcholine (suxamethonium) is classified as a: A) Non-depolarizing muscle relaxant B) Depolarizing muscle relaxant C) Local anaesthetic D) Reversal agent

Answer: B) Depolarizing muscle relaxant **Explanation:** Succinylcholine is the only depolarizing neuromuscular blocker in routine clinical use; it causes characteristic muscle fasciculations before paralysis and is broken down rapidly by plasma pseudocholinesterase.

Q51. A known dangerous side effect of succinylcholine in burns, major trauma, or neuromuscular disease patients is: A) Hypokalemia B) Hyperkalemia C) Hypoglycemia D) Hyponatremia

Answer: B) Hyperkalemia **Explanation:** In patients with denervation injury, burns, or prolonged immobilization, succinylcholine can trigger massive potassium efflux from upregulated extrajunctional receptors, causing dangerous hyperkalemia and cardiac arrest.

Q52. Atracurium is metabolized independent of renal or hepatic function via: A) Plasma cholinesterase hydrolysis B) Hofmann elimination C) Renal excretion only D) Hepatic conjugation only

Answer: B) Hofmann elimination **Explanation:** Atracurium undergoes spontaneous, non-enzymatic Hofmann elimination (a temperature/pH-dependent chemical breakdown), making it useful in patients with renal or hepatic impairment.

Q53. Among non-depolarizing muscle relaxants, which has the fastest onset, making it a useful alternative to succinylcholine for rapid sequence induction? A) Atracurium B) Vecuronium C) Rocuronium D) Pancuronium

Answer: C) Rocuronium **Explanation:** At an adequate intubating dose, rocuronium has the fastest onset among non-depolarizing agents (~60–90 seconds) and can be rapidly reversed by sugammadex, making it a common RSI alternative to succinylcholine.

Q54. Sugammadex is a selective reversal agent specifically for: A) Opioids B) Benzodiazepines C) Rocuronium and vecuronium D) Atracurium

Answer: C) Rocuronium and vecuronium **Explanation:** Sugammadex is a modified cyclodextrin that encapsulates the steroidal neuromuscular blockers rocuronium and vecuronium, rapidly terminating their effect; it is not effective against benzylisoquinolinium agents like atracurium.

Q55. Neostigmine, used to reverse non-depolarizing neuromuscular blockade, is typically co-administered with glycopyrrolate or atropine to: A) Increase its potency B) Counteract muscarinic side effects like bradycardia C) Speed up its metabolism D) Prevent allergic reactions

Answer: B) Counteract muscarinic side effects like bradycardia **Explanation:** Neostigmine non-selectively increases acetylcholine at both nicotinic (desired) and muscarinic (bradycardia, salivation, bronchospasm) receptors; an antimuscarinic blunts these unwanted effects.

Q56. Fentanyl is approximately how many times more potent than morphine? A) 10 times B) 100 times C) 2 times D) 1000 times

Answer: B) 100 times **Explanation:** Fentanyl is a highly lipid-soluble synthetic opioid roughly 100 times more potent than morphine by weight, with a faster onset and shorter duration of action.

Q57. The first sign of local anaesthetic systemic toxicity (LAST) is typically: A) Cardiac arrest B) Perioral tingling and tinnitus (CNS symptoms) C) Hypertension D) Bronchospasm

Answer: B) Perioral tingling and tinnitus (CNS symptoms) **Explanation:** LAST classically progresses from early CNS excitation (perioral numbness, tinnitus, metallic taste, agitation) to seizures, and only later to cardiovascular depression and arrhythmias if untreated.

Q58. Lipid emulsion ("Intralipid") therapy is specifically used in the management of: A) Malignant hyperthermia B) Local anaesthetic systemic toxicity C) Anaphylaxis D) Opioid overdose

Answer: B) Local anaesthetic systemic toxicity **Explanation:** IV lipid emulsion acts as a "lipid sink," sequestering lipophilic local anaesthetic molecules away from cardiac tissue, and is now a standard rescue therapy for severe LAST (e.g., bupivacaine cardiotoxicity).

Q59. The maximum safe dose of plain (without adrenaline) lignocaine is approximately: A) 1 mg/kg B) 3 mg/kg C) 7 mg/kg D) 10 mg/kg

Answer: B) 3 mg/kg **Explanation:** Plain lignocaine's maximum recommended dose is about 3 mg/kg; adding adrenaline (which reduces systemic absorption via vasoconstriction) raises the safe ceiling to about 7 mg/kg.

Q60. Compared to lignocaine, bupivacaine is: A) Shorter-acting and less cardiotoxic B) Longer-acting and more cardiotoxic C) Not used for spinal anaesthesia D) Only used topically

Answer: B) Longer-acting and more cardiotoxic **Explanation:** Bupivacaine's longer duration makes it popular for spinal/epidural anaesthesia and postoperative analgesia, but it carries a higher risk of severe, difficult-to-treat cardiotoxicity in overdose compared to lignocaine.

Q61. Spinal anaesthesia is conventionally administered at or below which vertebral interspace, to avoid injury to the spinal cord? A) T12–L1 B) L1–L2 C) L3–L4 D) C7–T1

Answer: C) L3–L4 **Explanation:** The spinal cord (conus medullaris) typically ends around L1–L2 in adults; needle insertion at L3–L4 or L4–L5, below this level, avoids direct cord trauma while still accessing the subarachnoid space.

Q62. The "loss of resistance" technique is used to identify the: A) Subarachnoid space for spinal anaesthesia B) Epidural space C) Brachial plexus D) Peripheral vein

Answer: B) Epidural space **Explanation:** As a Tuohy needle advances through the ligamentum flavum into the epidural space, resistance to a saline- or air-filled syringe suddenly disappears, signalling correct needle placement.

Q63. Dantrolene sodium is the specific treatment for: A) Anaphylaxis B) Malignant hyperthermia C) Local anaesthetic toxicity D) Opioid-induced respiratory depression

Answer: B) Malignant hyperthermia **Explanation:** Dantrolene inhibits calcium release from the sarcoplasmic reticulum via the ryanodine receptor, directly counteracting the uncontrolled muscle

contraction and hypermetabolism seen in malignant hyperthermia.

Q64. Flumazenil is the specific reversal agent for: A) Opioid overdose B) Benzodiazepine overdose/sedation C) Muscle relaxant overdose D) Local anaesthetic toxicity

Answer: B) Benzodiazepine overdose/sedation **Explanation:** Flumazenil is a competitive benzodiazepine receptor antagonist that reverses sedation from drugs like midazolam; its short half-life means re-sedation can occur, so monitoring is required after use.

Q65. Minimum Alveolar Concentration (MAC) is defined as the alveolar concentration of an inhalational agent that: A) Causes unconsciousness in 100% of patients B) Prevents movement in response to a surgical stimulus in 50% of patients C) Causes respiratory arrest D) Produces maximum muscle relaxation

Answer: B) Prevents movement in response to a surgical stimulus in 50% of patients **Explanation:** MAC (at 1 atmosphere) is the standard measure of inhalational anaesthetic potency; a lower MAC value indicates a more potent agent (e.g., halothane's MAC is much lower than desflurane's).

Section D — Patient Monitoring, Fluids & Emergency Management (Q66–Q85)

Q66. Pulse oximetry estimates oxygen saturation based on: A) Differential absorption of red and infrared light by oxygenated and deoxygenated haemoglobin B) Direct measurement of dissolved oxygen in plasma C) Measurement of exhaled CO₂ D) Electrical conductivity of blood

Answer: A) Differential absorption of red and infrared light by oxygenated and deoxygenated haemoglobin **Explanation:** Oxygenated and deoxygenated haemoglobin absorb red (660 nm) and infrared (940 nm) light differently; the pulse oximeter uses the ratio of pulsatile absorption at these wavelengths to calculate SpO₂.

Q67. Capnography is considered the gold standard for confirming: A) Depth of anaesthesia B) Correct tracheal (versus oesophageal) placement of an endotracheal tube C) Adequate muscle relaxation D) Blood pressure control

Answer: B) Correct tracheal (versus oesophageal) placement of an endotracheal tube **Explanation:** A sustained, characteristic square-wave CO₂ trace confirms the ETT is in the trachea, since the oesophagus does not deliver CO₂-containing gas, making capnography the most reliable real-time confirmation method.

Q68. The normal range for end-tidal CO₂ (EtCO₂) is approximately: A) 15–20 mmHg B) 35–45 mmHg C) 60–70 mmHg D) 80–90 mmHg

Answer: B) 35–45 mmHg **Explanation:** EtCO₂ of 35–45 mmHg closely approximates arterial PaCO₂ in patients with normal cardiopulmonary function, and is used to guide ventilation adequacy

intraoperatively.

Q69. Non-invasive blood pressure (NIBP) monitors used in the OT typically work on the principle of: A) Auscultation only B) Oscillometry C) Direct arterial cannulation D) Doppler ultrasound only

Answer: B) Oscillometry **Explanation:** Automated NIBP cuffs detect oscillations in cuff pressure caused by arterial pulsations as the cuff deflates, and algorithmically calculate systolic, diastolic, and mean arterial pressures.

Q70. Which ECG lead is most commonly monitored intraoperatively for early detection of myocardial ischaemia? A) Lead I B) Lead aVR C) Lead II D) Lead V5

Answer: D) Lead V5 **Explanation:** Lead V5 overlies the left ventricular lateral wall and detects most intraoperative ischaemic episodes; Lead II is monitored separately for its clear P-wave visualization and arrhythmia detection.

Q71. The normal central venous pressure (CVP), with the catheter tip at the superior vena cava–right atrial junction, is approximately: A) 2–8 mmHg B) 20–25 mmHg C) 40–50 mmHg D) -5 to 0 mmHg

Answer: A) 2–8 mmHg **Explanation:** A normal CVP of roughly 2–8 mmHg reflects right ventricular preload and is used, alongside other parameters, to guide fluid management in critically ill or major surgical patients.

Q72. Ringer Lactate, unlike normal saline, additionally contains: A) Potassium, calcium, and lactate B) Only glucose C) Only bicarbonate D) Magnesium sulfate

Answer: A) Potassium, calcium, and lactate **Explanation:** Ringer Lactate more closely resembles plasma composition than 0.9% normal saline, adding potassium, calcium, and lactate (which is metabolized to bicarbonate in the liver, providing a buffering effect).

Q73. 5% Dextrose (D5) solution, though isotonic in the infusion bag, behaves in vivo as: A) A hypertonic solution B) An effectively hypotonic (free-water) solution once glucose is metabolized C) A colloid D) A pure isotonic solution with no free-water effect

Answer: B) An effectively hypotonic (free-water) solution once glucose is metabolized **Explanation:** Once the dextrose in D5W is metabolized, only free water remains, distributing across all body fluid compartments — so despite being isotonic on paper, its physiological effect is that of a hypotonic fluid.

Q74. Using the Holliday-Segar (4-2-1) formula, the maintenance IV fluid rate for a 25 kg child is: A) 25 mL/hr B) 50 mL/hr C) 65 mL/hr D) 100 mL/hr

Answer: C) 65 mL/hr **Explanation:** The 4-2-1 rule gives 4 mL/kg/hr for the first 10 kg (40) + 2 mL/kg/hr for the next 10 kg (20) + 1 mL/kg/hr for each kg above 20 kg (5 for the remaining 5 kg) = 65 mL/hr.

Q75. The universal blood donor group (red cell component) is: A) AB positive B) O negative C) A positive D) B negative

Answer: B) O negative **Explanation:** O negative red cells lack A, B, and Rh(D) antigens, making them least likely to trigger a transfusion reaction in a recipient of unknown blood group — hence the "universal donor" in emergencies.

Q76. During adult single-rescuer CPR, the recommended compression-to-ventilation ratio is: A) 15:2 B) 30:2 C) 5:1 D) 10:2

Answer: B) 30:2 **Explanation:** Current adult BLS guidelines recommend 30 chest compressions to 2 rescue breaths for single-rescuer CPR, with compressions delivered at 100–120/min and a depth of at least 5 cm.

Q77. Which cardiac arrest rhythms are considered "shockable" and treated with immediate defibrillation? A) Asystole and PEA B) Ventricular fibrillation and pulseless ventricular tachycardia C) Sinus bradycardia and first-degree AV block D) Atrial fibrillation only

Answer: B) Ventricular fibrillation and pulseless ventricular tachycardia **Explanation:** VF and pulseless VT are chaotic or excessively rapid rhythms that respond to defibrillation; asystole and PEA are "non-shockable" rhythms managed instead with high-quality CPR and drugs like adrenaline.

Q78. Perioperative hypothermia in surgical patients is primarily caused by: A) High OT temperature B) Anaesthesia-induced impairment of thermoregulation combined with a cold OT environment and exposed body cavities C) Excessive patient movement D) High humidity only

Answer: B) Anaesthesia-induced impairment of thermoregulation combined with a cold OT environment and exposed body cavities **Explanation:** General and regional anaesthesia blunt normal vasoconstrictive/shivering responses, so combined with a cool OT, cold IV fluids, and open cavities, patients lose core heat rapidly unless actively warmed.

Q79. The Aldrete scoring system is used to assess a patient's readiness for: A) Induction of anaesthesia B) Discharge from the post-anaesthesia care unit (PACU) C) Extubation on the operating table D) ICU admission criteria

Answer: B) Discharge from the post-anaesthesia care unit (PACU) **Explanation:** The Aldrete score evaluates activity, respiration, circulation, consciousness, and oxygen saturation; a score of 9–10 generally indicates the patient is fit to be transferred out of the recovery area.

Q80. To prevent diffusion hypoxia after discontinuing nitrous oxide, the recommended practice is to: A) Immediately extubate the patient B) Administer 100% oxygen for a few minutes C) Give an antiemetic D) Increase nitrous oxide flow temporarily

Answer: B) Administer 100% oxygen for a few minutes **Explanation:** When N₂O is stopped, it diffuses rapidly out of the blood into the alveoli, transiently diluting alveolar oxygen ("diffusion hypoxia"); giving 100% oxygen for a few minutes prevents hypoxia during this washout period.

Q81. A size "E" oxygen cylinder, commonly mounted on anaesthesia/transport machines, has an approximate capacity of: A) 100 litres B) 300 litres C) 680 litres D) 2000 litres

Answer: C) 680 litres **Explanation:** A full size-E oxygen cylinder (filled to ~137 bar/13,700 kPa) holds approximately 680 litres of gas — a useful figure for technicians to mentally estimate how long a cylinder will last at a given flow rate.

Q82. In massive blood transfusion protocols, blood products are typically administered in a balanced ratio of packed red cells, fresh frozen plasma, and platelets of approximately: A) 1:1:1 B) 5:1:1 C) 1:5:1 D) 10:1:1

Answer: A) 1:1:1 **Explanation:** A roughly 1:1:1 ratio of PRBC:FFP:platelets mimics whole blood composition and is associated with better outcomes in massive haemorrhage compared to crystalloid-heavy or red-cell-only resuscitation.

Q83. The APGAR score is formally assessed at: A) Only at birth B) 1 and 5 minutes after birth C) 10 and 20 minutes after birth D) Every hour for the first day

Answer: B) 1 and 5 minutes after birth **Explanation:** The APGAR score is conventionally recorded at 1 minute (reflecting need for immediate resuscitation) and again at 5 minutes (reflecting response to resuscitation); it may be repeated at 10 minutes if the score remains low.

Q84. A steadily falling oxygen cylinder pressure gauge reading, at a fixed flow rate, indicates: A) The cylinder is overfilled B) The cylinder is progressively emptying C) The flowmeter is faulty D) The vaporizer needs refilling

Answer: B) The cylinder is progressively emptying **Explanation:** Because oxygen is stored as a compressed gas, cylinder pressure falls roughly in proportion to remaining content, so a technician can estimate remaining supply duration from the pressure gauge reading and current flow rate.

Q85. Which of the following best describes the primary purpose of a difficult airway cart in the OT complex? A) Storage of routine surgical instruments B) Ready access to specialized equipment (video laryngoscope, LMAs, bougies, cricothyroidotomy kit) for anticipated or unanticipated difficult intubation C) Storage of IV fluids only D) Storage of sterilization chemicals

Answer: B) Ready access to specialized equipment (video laryngoscope, LMAs, bougies, cricothyroidotomy kit) for anticipated or unanticipated difficult intubation **Explanation:** A well-

stocked, immediately accessible difficult airway cart is a key patient-safety measure, allowing rapid escalation through airway rescue devices when standard laryngoscopy and intubation fail.

Section E — Surgical Procedures, Positions & Terminology (Q86–Q100)

Q86. The Trendelenburg position (head-down tilt) is commonly used during: A) Neurosurgical procedures B) Lower abdominal and pelvic laparoscopic surgery C) Upper limb surgery D) ENT surgery only

Answer: B) Lower abdominal and pelvic laparoscopic surgery **Explanation:** Tilting the head down shifts abdominal contents cephalad, improving surgical access to the pelvis during gynaecological and lower abdominal laparoscopic procedures.

Q87. The lithotomy position, with the patient's legs elevated and supported in stirrups, is most commonly used for: A) Craniotomy B) Gynaecological, urological, and rectal procedures C) Shoulder surgery D) Spine surgery

Answer: B) Gynaecological, urological, and rectal procedures **Explanation:** Lithotomy position provides surgical access to the perineum and pelvis and is standard for procedures like vaginal hysterectomy, cystoscopy, and haemorrhoidectomy, but requires careful padding to avoid nerve injury.

Q88. The prone position is most typically required for: A) Laparoscopic cholecystectomy B) Posterior spine surgery C) Thyroidectomy D) Cataract surgery

Answer: B) Posterior spine surgery **Explanation:** Prone positioning provides direct access to the posterior spine and posterior fossa, but requires careful attention to eye protection, chest/abdominal support, and airway security since the face is not visible to the anaesthetist.

Q89. McBurney's incision, made at McBurney's point, is the classical open surgical approach for: A) Cholecystectomy B) Appendectomy C) Herniorrhaphy D) Nephrectomy

Answer: B) Appendectomy **Explanation:** McBurney's point lies at the junction of the lateral one-third and medial two-thirds of a line from the anterior superior iliac spine to the umbilicus, overlying the base of the appendix in most patients.

Q90. Which surgical procedure involves excision of a hernial sac only, without repair of the underlying defect, mainly used in paediatric/congenital hernias? A) Herniorrhaphy B) Hernioplasty C) Herniotomy D) Herniectomy

Answer: C) Herniotomy **Explanation:** Herniotomy (sac excision/ligation alone) suffices in congenital hernias where the abdominal wall defect is expected to close with growth, unlike adult hernias which typically also require herniorrhaphy (suture repair) or hernioplasty (mesh repair).

Q91. Tracheostomy is conventionally performed between which tracheal rings? A) 1st–2nd B) 2nd–4th C) 6th–8th D) At the cricoid cartilage itself

Answer: B) 2nd–4th **Explanation:** A tracheostomy is placed through the 2nd–4th tracheal rings, avoiding the 1st ring and cricoid cartilage (which risks subglottic stenosis) and staying above the innominate artery crossing lower in the neck.

Q92. Stapedectomy, performed for otosclerosis, involves: A) Removal of the tympanic membrane B) Replacement of the stapes bone with a prosthesis C) Drainage of middle ear fluid D) Removal of the tonsils

Answer: B) Replacement of the stapes bone with a prosthesis **Explanation:** In otosclerosis, abnormal bone growth fixes the stapes footplate, impairing sound transmission; stapedectomy removes the fixed stapes and replaces it with a mobile prosthesis to restore hearing.

Q93. The Boyle-Davis mouth gag is a self-retaining instrument classically used during: A) Tonsillectomy B) Cataract surgery C) Hernia repair D) Cardiac surgery

Answer: A) Tonsillectomy **Explanation:** The Boyle-Davis gag holds the mouth open and depresses the tongue, giving the surgeon exposed access to the tonsillar fossae, while also serving as a conduit to secure the endotracheal tube in the midline.

Q94. Phacoemulsification is the modern standard surgical technique for: A) Cataract removal B) Retinal detachment repair C) Glaucoma surgery D) Corneal transplant

Answer: A) Cataract removal **Explanation:** Phacoemulsification uses ultrasonic energy delivered through a small incision to fragment and aspirate the cataractous lens, which is then typically replaced with a folded intraocular lens implant.

Q95. TURP syndrome, a recognized complication of transurethral resection of the prostate, is primarily caused by: A) Excessive blood loss only B) Absorption of large volumes of hypotonic irrigation fluid, causing dilutional hyponatremia C) Allergic reaction to anaesthesia D) Nerve injury from the lithotomy position

Answer: B) Absorption of large volumes of hypotonic irrigation fluid, causing dilutional hyponatremia **Explanation:** During TURP, opened venous sinuses can absorb large volumes of hypotonic irrigant (classically 1.5% glycine), causing fluid overload, dilutional hyponatremia, and glycine-related toxicity presenting as confusion, visual disturbance, and cardiovascular instability.

Q96. In laparoscopic surgery, carbon dioxide is the preferred gas for creating pneumoperitoneum mainly because it is: A) Flammable and cheap B) Non-flammable and highly soluble in blood, reducing the severity of gas embolism C) Odourless only D) The only gas that supports combustion

Answer: B) Non-flammable and highly soluble in blood, reducing the severity of gas embolism **Explanation:** Unlike air, oxygen, or nitrous oxide, CO₂ does not support combustion during electrocautery use and is rapidly absorbed/excreted by the lungs if it enters the bloodstream, making accidental gas embolism less catastrophic.

Q97. The intra-abdominal pressure typically maintained during laparoscopic surgery is approximately: A) 2–5 mmHg B) 12–15 mmHg C) 25–30 mmHg D) 40–50 mmHg

Answer: B) 12–15 mmHg **Explanation:** An insufflation pressure of roughly 12–15 mmHg provides adequate surgical working space while limiting adverse effects on venous return, diaphragmatic excursion, and cardiac output.

Q98. The Veress needle is primarily used to: A) Suture skin B) Create initial pneumoperitoneum before trocar insertion in laparoscopic surgery C) Administer spinal anaesthesia D) Insert a central line

Answer: B) Create initial pneumoperitoneum before trocar insertion in laparoscopic surgery **Explanation:** The spring-loaded, blunt-tipped Veress needle is inserted through a small incision to safely insufflate CO₂ into the peritoneal cavity before the first trocar is placed, minimizing risk of visceral injury.

Q99. Which of the following is the classical open surgical incision for cholecystectomy? A) McBurney's incision B) Kocher's (subcostal) incision C) Pfannenstiel incision D) Midline sternotomy

Answer: B) Kocher's (subcostal) incision **Explanation:** Kocher's incision, a right subcostal incision paralleling the costal margin, provides direct access to the gallbladder and biliary tree in open cholecystectomy, largely superseded today by the four-port laparoscopic approach.

Q100. The lateral (kidney) position, often with the operating table flexed and a kidney rest elevated, is primarily used for: A) Renal and retroperitoneal surgery B) Cardiac bypass surgery C) Caesarean section D) Thyroid surgery

Answer: A) Renal and retroperitoneal surgery **Explanation:** The lateral position with flexion opens the space between the costal margin and iliac crest, improving surgical access to the kidney and retroperitoneum, though it requires careful axillary roll placement to prevent brachial plexus injury and dependent-lung compression.

Quick-Reference Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	21	B	41	B	61	C	81	C
2	B	22	B	42	B	62	B	82	A

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
3	B	23	B	43	A	63	B	83	B
4	A	24	B	44	A	64	B	84	B
5	B	25	B	45	B	65	B	85	B
6	B	26	D	46	C	66	A	86	B
7	A	27	B	47	B	67	B	87	B
8	A	28	C	48	C	68	B	88	B
9	A	29	B	49	B	69	B	89	B
10	B	30	B	50	B	70	D	90	C
11	B	31	A	51	B	71	A	91	B
12	C	32	D	52	B	72	A	92	B
13	B	33	D	53	C	73	B	93	A
14	B	34	B	54	C	74	C	94	A
15	B	35	A	55	B	75	B	95	B
16	A	36	B	56	B	76	B	96	B
17	C	37	B	57	B	77	B	97	B
18	B	38	A	58	B	78	B	98	B
19	B	39	B	59	B	79	B	99	B
20	B	40	C	60	B	80	B	100	A

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